

Chapter 6

Financial markets and the extended IS-LM model

What is this chapter about?

- Assumption so far: two financial assets - money and bonds - and one interest rate - the rate on bonds - determined by monetary policy
- We now relax this assumption
- Why? Because the financial system plays a major role in the economy, and we want to study its macroeconomics implications

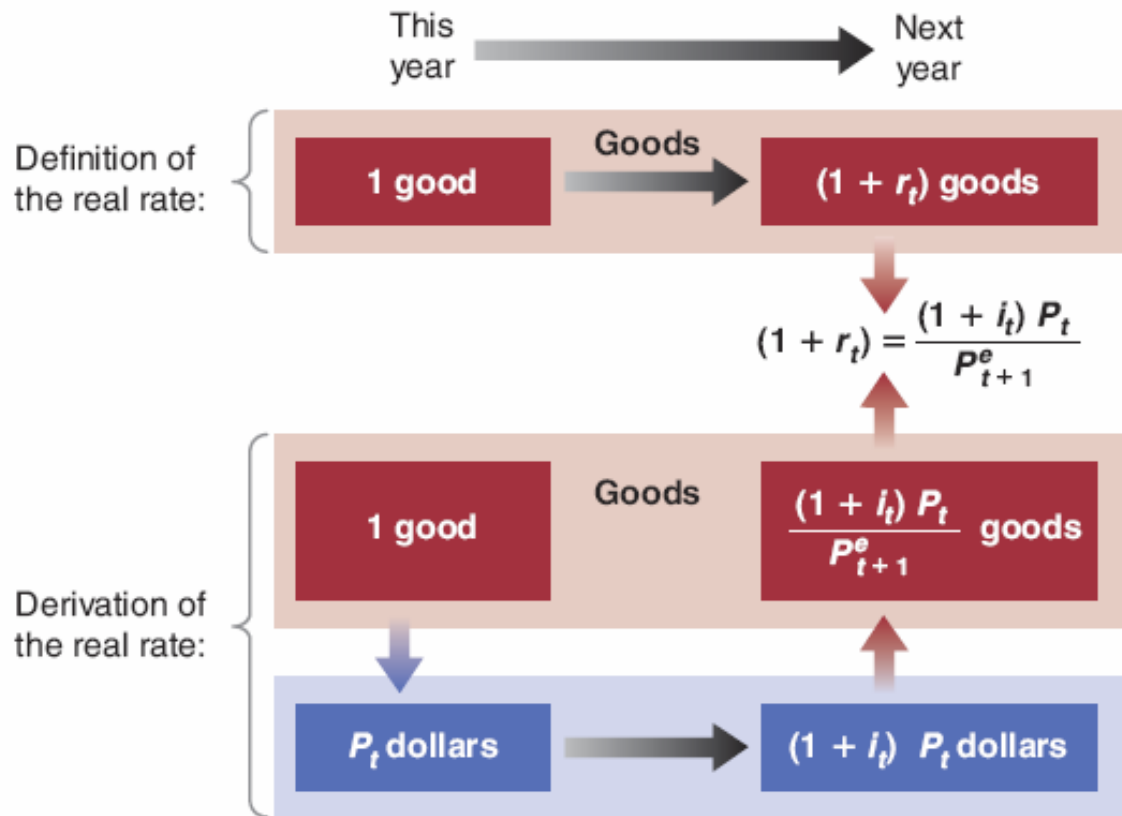
Outline

- Nominal versus real interest rates
- Risk and risk premiums
- Extending the IS-LM to think about financial shocks

Nominal versus real interest rates

- **Nominal interest rate:** interest rate in terms of dollars/euros
- **Real interest rate:** interest rate in terms of a basket of goods
- To obtain the real interest rate, we must adjust the nominal interest rate for **expected inflation**

Definition and derivation of the real interest rate



- We derived the following expression for the **one-year real interest rate**

$$1 + r_t = \frac{(1 + i_t)P_t}{P_{t+1}^e}$$

- Now, note that **expected inflation** between t and $t + 1$ is

$$\pi_{t+1}^e = \frac{P_{t+1}^e - P_t}{P_t} \quad \text{or equivalently} \quad 1 + \pi_{t+1}^e = \frac{P_{t+1}^e}{P_t}$$

- **Substituting**, we can write the real rate as

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e}$$

- Finally, for small i and π^e , the **previous expression can be approximated** by

$$r_t \approx i_t - \pi_{t+1}^e$$

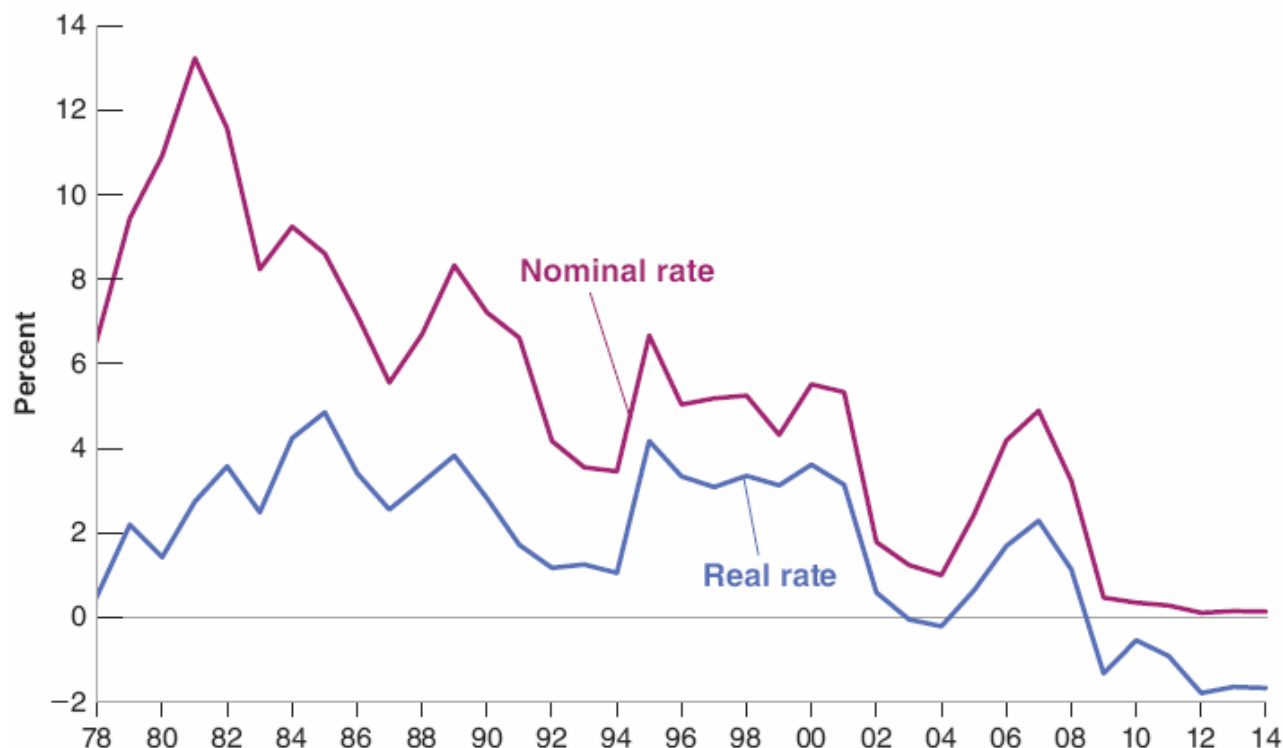
$$r_t \approx i_t - \pi_{t+1}^e$$

- What does this equation tell us?
 - When expected inflation equals zero, the nominal interest rate and the real interest rate are equal
 - Because expected inflation is typically positive, the real interest rate is typically lower than the nominal interest rate
 - For a given nominal interest rate, the higher expected inflation, the lower the real interest rate

- The **real interest rate**, $r = i - \pi^e$, is based on expected inflation, so it is sometimes called the **ex-ante real interest rate**
- The **realized real interest rate**, $r = i - \pi$, is called the **ex-post interest rate**

Nominal and real T-Bill rates, US, 1978-2014

Source: Nominal Interest rate is the 1-year Treasury bill in December of the previous year: Series TB1YR, Federal Reserve Economic Data (FRED) <http://research.stlouisfed.org/fred2/> (Series TB6MS in December 2001, 2002, 2003, and 2004.) Expected Inflation is the 12-month forecast of inflation, using the GDP deflator, from the December OECD Economic Outlook from the previous year.



The nominal interest rate has declined considerably since early 1980s, but since expected inflation has declined as well, the real rate has declined much less than the nominal rate

Real or nominal interest rate in the IS?

- What interest rate should enter the IS relation? The **real rate**!
- Why? What matters for consumption and investment decisions is the **real interest rate**

What are the implications for monetary policy?

- While the **CB sets the nominal interest rate**, it **cares about the real interest rate**, as it is the one affecting spending decisions
- To set the real rate it wants, the CB has to account for expected inflation, and choose the nominal rate so that, given expected inflation, $r = i - \pi^e$
- For example, if the CB wants $r = 4\%$ and $\pi^e = 2\%$, then it sets $i = 6\%$
- Note that the **zero lower bound of the nominal interest rate** implies that the real interest rate cannot be lower than the negative of inflation
- If $\pi^e = 2\%$, the real interest rate cannot be lower than $0 - 2\% = -2\%$
If $\pi^e = -1\%$, real interest rate cannot be lower than $0 - (-1\%) = 1\%$

Risk and risk premiums

- **Some bonds are risky**, since there is a non negligible probability that the borrower will not be able or willing to repay
- To compensate for the risk, bond holders require a **risk premium**
- What factors determine risk premia?
 1. **Probability of default** of the borrower
 2. **Degree of risk aversion** of the bond holders

1. Probability of default and risk premiums

Let i be the nominal interest rate on a riskless bond, x be the risk premium, and p is the probability of defaulting, then to get the same expected return on the risky bond as on the riskless bond, it has to be that

$$(1 + i) = (1 - p) (1 + i + x) + p (0)$$

so that

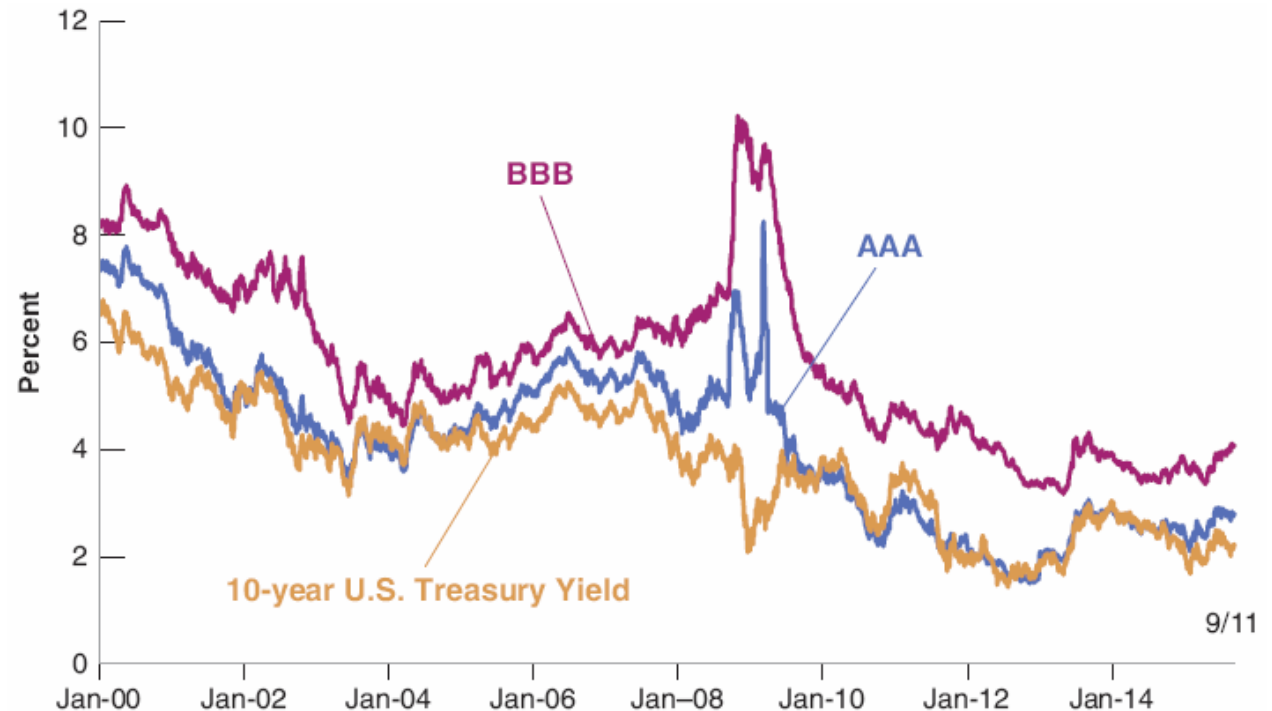
$$x = (1 + i) p / (1 - p)$$

2. Risk aversion and risk premia

- Even if the expected return on risky bonds is the same than the return on riskless bonds, the risk itself may make bond holders reluctant to hold risky bonds
- Bond holders will require a higher risk premium, the higher is their degree of risk aversion, for given probability of default

Yields on 10-Year US Government Treasury, AAA and BBB Corporate Bonds, 2000-2014

Source: For AAA and BBB corporate bonds, Bank of America Merrill Lynch; for 10-year U.S. treasury yield, Federal Reserve Board.



During 2008 and 2009, while the rate on government bonds decreased, reflecting the decision of the Fed to decrease the policy rate, the financial crisis led to a sharp increase in the rates at which firms could borrow, especially for lower-rated bonds

Extending the IS-LM model

- Now we extend the IS-LM model to reflect the distinction between
 - the **nominal rate** i and the **real rate** r
 - the **policy rate** i set by the central bank and the interest rates faced by consumers and firms, the **borrowing rate** $r + x = i - \pi^e + x$
- Rewrite the IS-LM model as follows

$$IS \text{ relation : } Y = C(Y - T) + I(Y, i - \pi^e + x) + G$$

$$LM \text{ relation : } i = \bar{i}$$

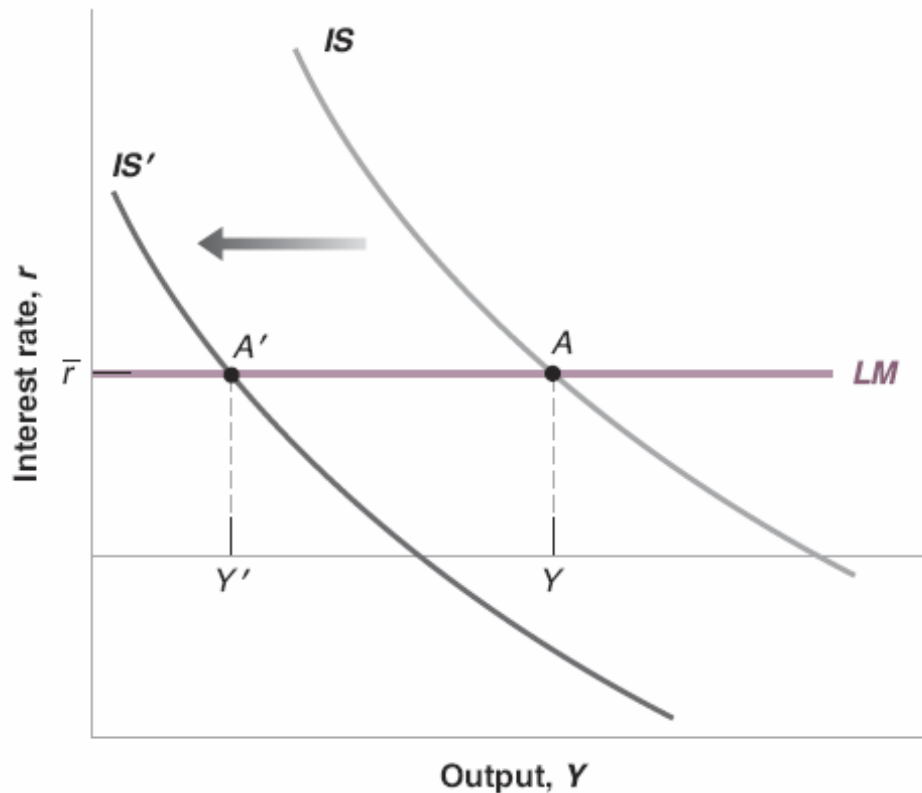
- A **higher risk premium** x captures in a simple way several factors:
 - **higher probability of defaulting**
 - **higher degree of risk aversion of lenders**
 - **lower lending from financial intermediaries due to solvency or liquidity worries**

- For simplicity and *ignoring issues associated to the zero lower bound*, we can think of the **CB choosing directly** the **real policy rate r** , which in turn **enters the IS equation** as part of the **borrowing rate $r + x$**
- The IS-LM model can be written as

$$IS \text{ relation : } Y = C(Y - T) + I(Y, r + x) + G$$

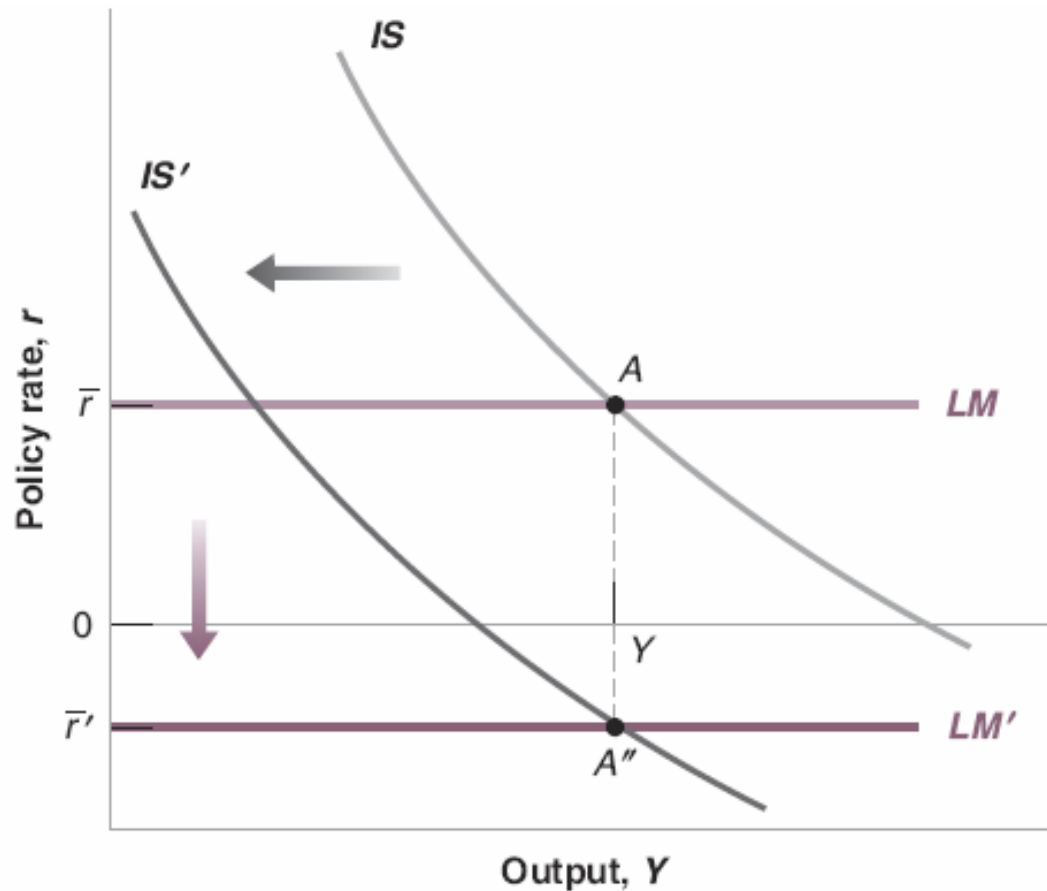
$$LM \text{ relation : } r = \bar{r}$$

Financial shocks and output



- An increase in x leads to a shift of the IS curve to the left and a decrease in equilibrium output
- An increase in x , given r , increases the borrowing rate, reduces investment, demand and output
- A financial crisis becomes a macroeconomic crisis (that is, it leads to a recession)

Financial shocks, monetary policy and output



- What can monetary policy do?
- If sufficiently large, a decrease in the policy rate can in principle offset the increase in the risk premium
- The *zero lower bound* may however put a limit on the decrease in the real policy rate